

CO4-AT3-Q2:

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Course Name: DAA for Complexity Analysis

Dice Throw Problem vs Coin Change (DP Comparison)

Problem Understanding

Dice Throw and Coin Change problems are Dynamic Programming problems. Both use stored previous results to avoid repeated calculations.

Dice Throw Problem

The objective is to find the number of ways to get a target sum using given dice.

State:

`dp[dice][sum]`

Represents number of ways using dice to achieve a particular sum.

Coin Change Problem

The objective is to find the number of ways or minimum coins required to make a target value.

State:

`dp[amount]`

Stores previous calculated solutions.

Similarities

- Both use Dynamic Programming.
- Both contain overlapping subproblems.
- Both use recurrence relations.
- Previous results are reused.

Differences

Feature	Dice Throw	Coin Change
Input	Dice count and target sum	Coins and amount

Feature	Dice Throw	Coin Change
State	dice + sum	amount
Choices	1 to 6 faces	Available coins
Constraint	Fixed dice sides	Variable coins

Complexity Analysis

Dice Throw

Time Complexity:

$O(\text{dice} \times \text{sum} \times 6)$

Space Complexity:

$O(\text{dice} \times \text{sum})$

Coin Change

Time Complexity:

$O(\text{amount} \times \text{coins})$

Space Complexity:

$O(\text{amount})$

Result Analysis

Dynamic Programming reduces repeated calculations and improves performance compared to recursive solutions.

Conclusion

Dice Throw and Coin Change demonstrate effective DP modeling techniques. Proper selection of states and recurrence relations helps solve large optimization problems efficiently.